

Pull-Ups

YC248-JR-074K7L

NOTE: !ALERT, DOES NOT RECEIVE A DEDICATED PULL-UP RESISTOR. THEREFOR AN INPUT PULL-UP OF THE ESP32 IS REQUIRED.

Sensor section

SCL_1<
SDA_1<
!INT_HRD
!INT_RTCD
!INT_ALSD
!INT_HD
!ALERTD

Date: sensor_section.kicad_sch

Power section

USB_DETECTD
BAT_STATD
CTRL_PWR_NPM

Date: power_section.kicad_sch

Logic section

SCL_1
SDA_1
!INT_HR
!INT_RTC
!INT_ALS
!INT_H
!ALERT
USB_DETECT
BAT_STAT
D+<
D-<
CTRL_PWR_NPM

Date: logic_section.kicad_sch

Peripherie

DOUT_NPM<
BTN<
BUZZERD
LEVER_LEFT<
LEVER_PUSH<
LEVER_RIGHT<
CS_SDD
CS_TFTD
DCD
MISO<
MOSID
PWM_TFTD
RSTD
SCKD

DIN_NPM
BTN
BUZZER
LEVER_LEFT
LEVER_PUSH
LEVER_RIGHT
CS_SD
CS_TFT
DC
MISO
MOSI
PWM_TFT
RST
SCK

Date: peripherie.kicad_sch

IF NOT SPECIFEID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /
File: NX-1SD.kicad_sch

Title:

Size: A5	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 1/5

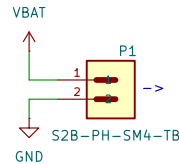
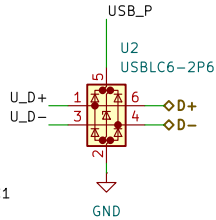
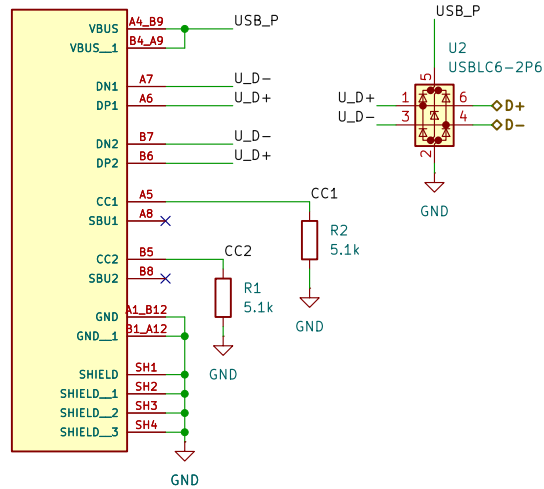
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Size: A5	
KiCad E.D.A. 10.0.5	

Rev:
Id: 1/5

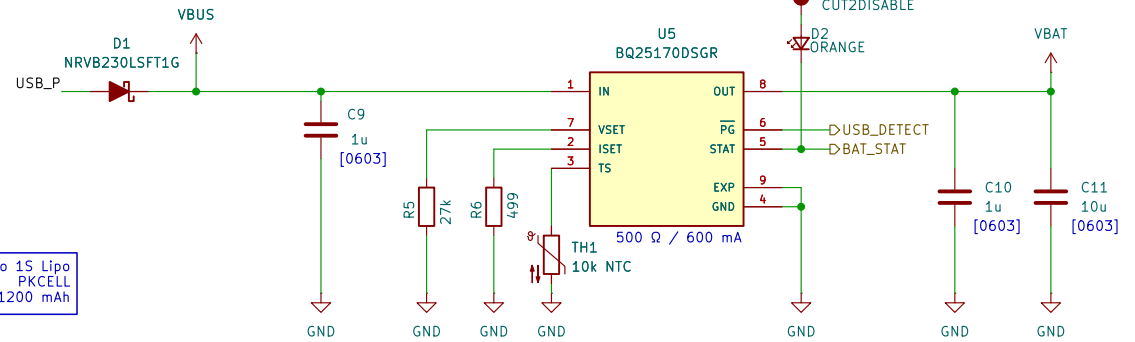
CONNECTORS

J1
UJ20-C-H-G-SMT-1-P16-TR



Connects to 1S Lipo
PKCELL
3.7 V, 1200 mAh

LIPO CHARGER

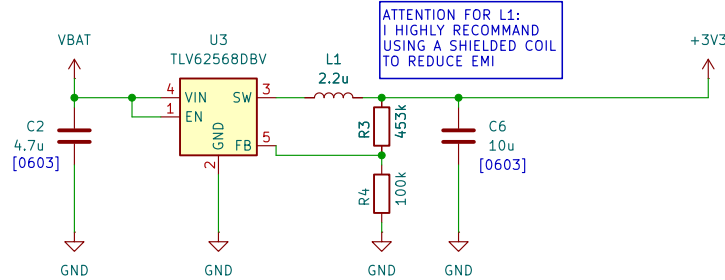


Formula $R_{ISET} = 300 / I_{OUT}(\text{in Amperes})$

Charge Current	Calculated R_{ISET}
50 mA	6000 Ω
100 mA	3000 Ω
150 mA	2000 Ω
200 mA	1500 Ω
300 mA	1000 Ω
400 mA	750 Ω
500 mA	600 Ω
600 mA	500 Ω
800 mA	375 Ω

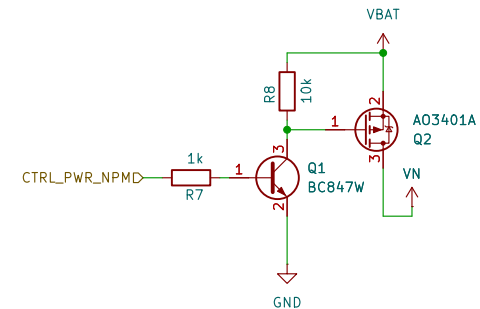
+3V3 DCDC Buck

DCDC: 3.3V / 1A



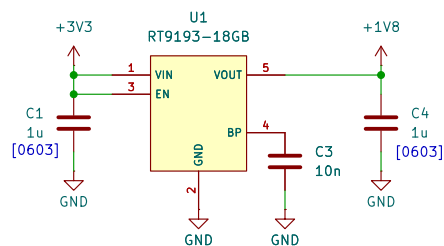
ATTENTION FOR L1:
I HIGHLY RECOMMEND
USING A SHIELDED COIL
TO REDUCE EMI

NPM POWER CONTROL



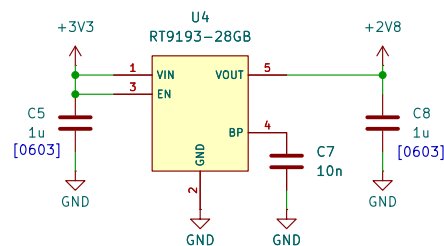
+1V8 LDO

(USED FOR HRM)



+2V8 LDO

(USED FOR DISPLAY)



IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Power section/
File: power_section.kicad_sch

Title:

Size: A4

Date:

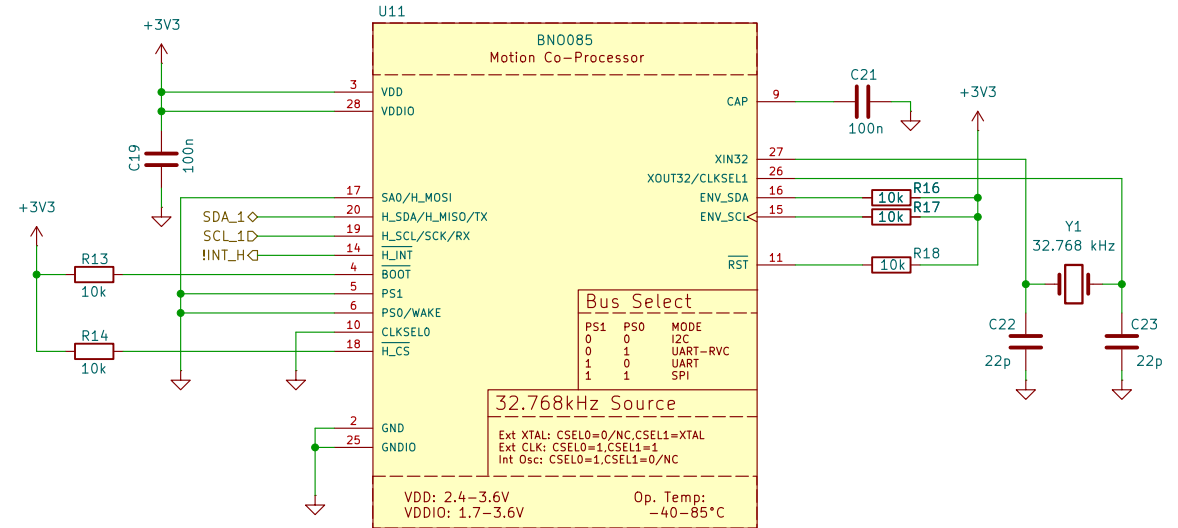
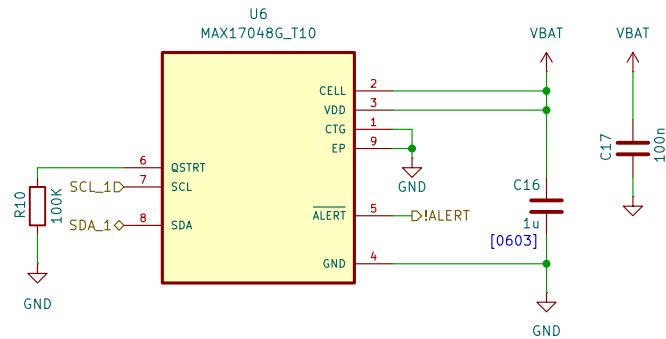
KiCad E.D.A. 10.0.5

Rev:

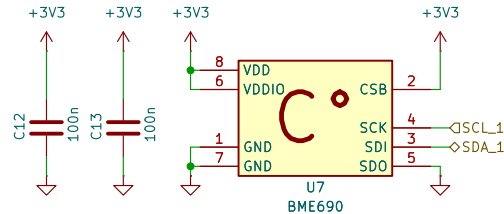
Id: 2/5

9-DOF

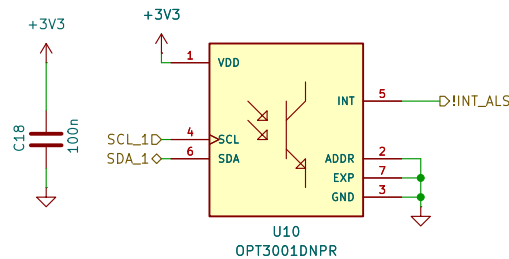
Battery Fuel Gauge



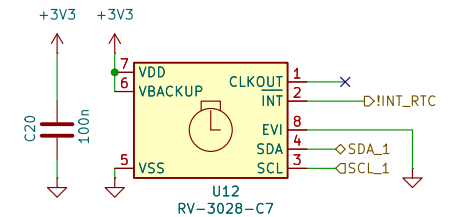
Temp/Humidity/Pressure



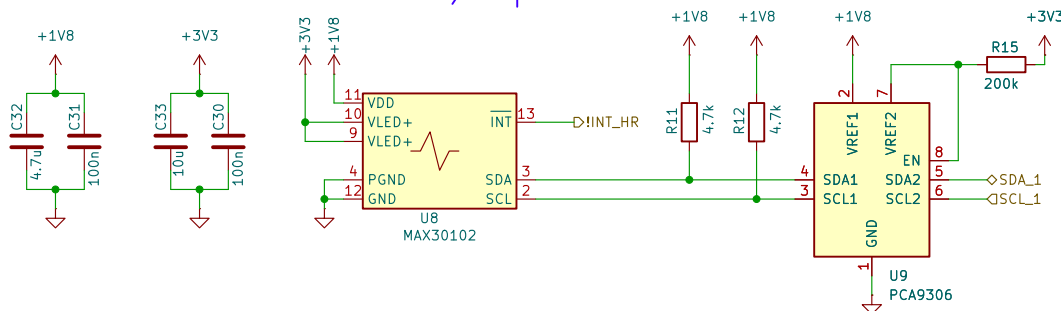
Ambient Light Sensor



RTC



BPM/SpO2



IF NOT SPECIFD IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Sensor section/
File: sensor_section.kicad_sch

Title:

Size: A4

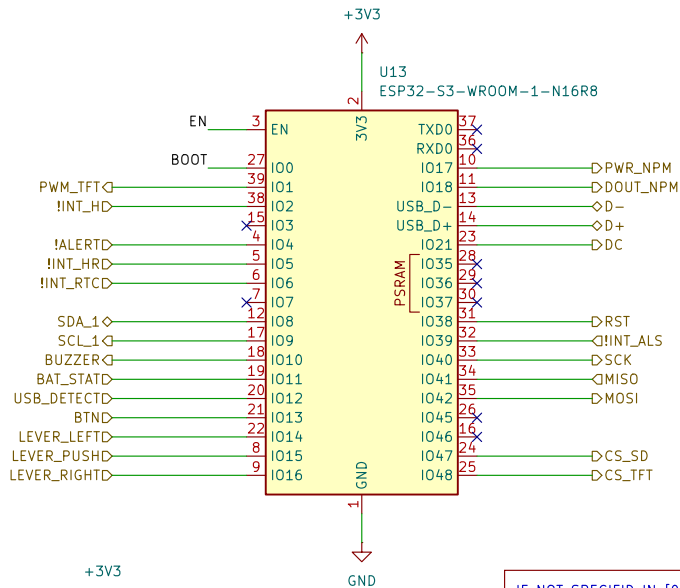
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KiCad E.D.A. 10.0.5

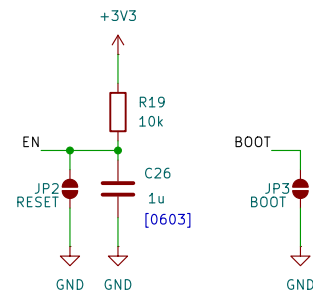
Rev:

Id: 3/5

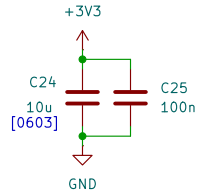
ESP32-S3 N16R8



EN & BOOT



ESP32-S3 chip strapping pins:
GPIO 0.
GPIO 3.
GPIO 45.
GPIO 46.



IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Logic section/
File: logic_section.kicad_sch

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Size: A5
KiCad E.D.A. 10.0.5

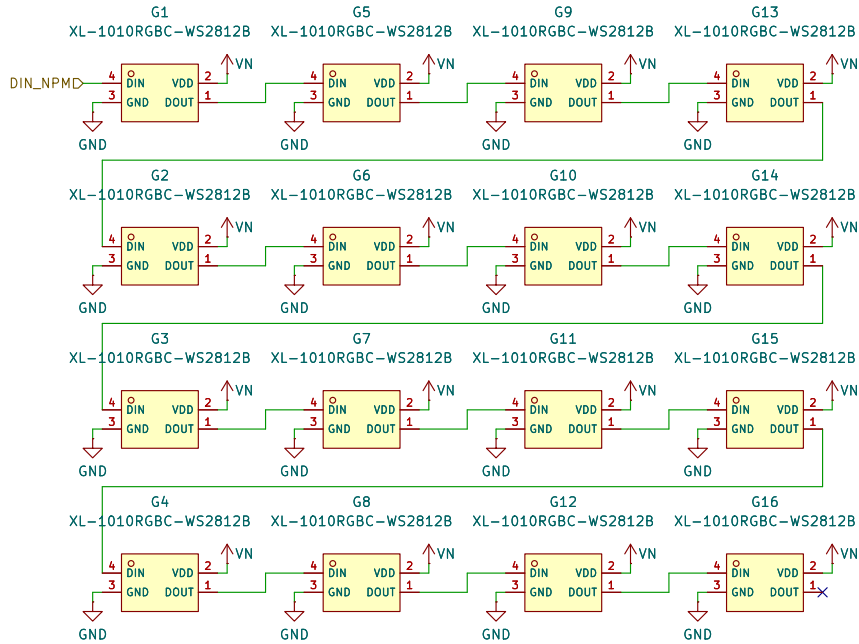
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Rev:

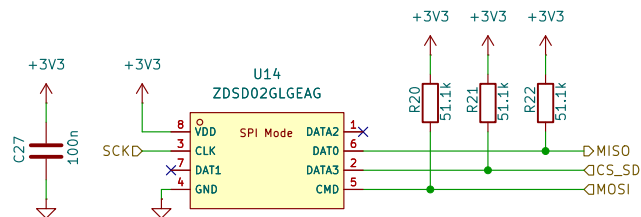
Id: 4/5

NeoPixel Matrix (NPM)

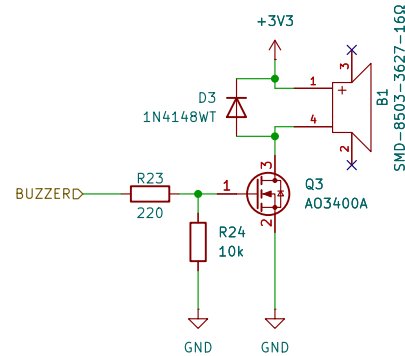
(NPM DOES NOT REQUIRE DECOUPLING CAPACITORS DUE TO THE SHORT PATH OF ITS SUPPLY)



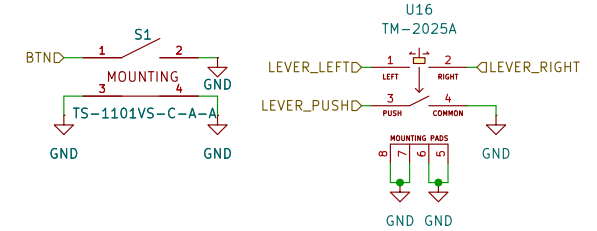
NAND-SD



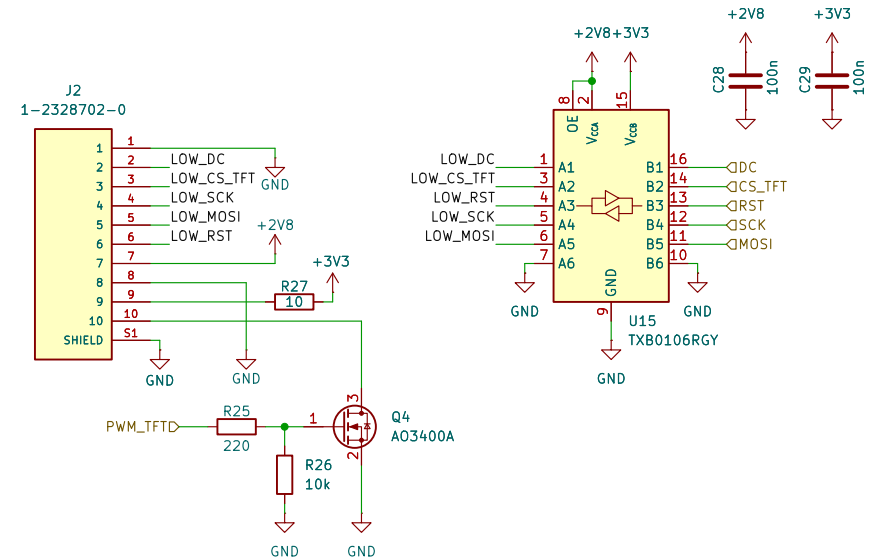
BUZZER



BUTTONS



DISPLAY



Sheet: /Peripherie/
File: peripherie.kicad_sch

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KiCad E.D.A. 10.0.5

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Id: 5/5